IDA117V | INTRODUCTION TO DATA SCIENCE

Predicting Household Water Supply Interruptions in South Africa

Group: [insert] | Members and student numbers: [insert] | Lecturer: [insert] | Date: [insert]

Technical report draft — evidence and scores must be populated from the included reproducible experiment outputs. No experimental metrics are invented.

1. Problem definition, stakeholder, data and ethics (Question 1)

Water service interruptions can undermine the reliability of household access to water. The affected population is South African households represented in the 2023 General Household Survey. Municipal water services managers and provincial oversight bodies are potential users of aggregate insights. The objective is to classify whether a household reported an interruption during the previous twelve months. This is retrospective classification, not an operational forecast of future outages.

Prediction unit: an eligible surveyed household. Target: wat_inte_12mth, with 1. Yes = interruption and 2. No = no interruption; 8. Not applicable is excluded. Inputs: province, geography, housing type and tenure, drinking-water source, sanitation, electricity access, refuse service, household head age and sex, household size and main income source. All inputs must be confirmed against the survey codebook before interpretation.

Primary metric: macro-F1, which gives each class equal weight. Secondary measures: recall of interruption cases, PR-AUC, ROC-AUC, accuracy and weighted-F1. A false negative could overlook households reporting unreliable supply; a false positive could direct attention to a household not reporting interruptions. Service indicators for any later field pilot: share of confirmed interruption cases identified, investigation turnaround time, and equitable geographic coverage. These are proposed indicators, not observed impacts.

Provenance: Statistics South Africa, General Household Survey 2023 household-level dataset, supplied CSV (20,927 rows and 188 columns). Source route listed in the assignment: https://www.statssa.gov.za/?p=17836. The exact download page, licensing terms, survey methodology and data dictionary require confirmation before submission. Risks include POPIA compliance, re-identification through linked microdata, biased coverage, spurious regional correlations and use of household risk labels to exclude residents. Safeguards: no identifying columns in the model, local processing, aggregation for reporting, access controls, fairness subgroup review and human oversight.

2. Audit, exploratory analysis and leakage-safe preparation (Question 2)

The original CSV contains 20,927 household records and 188 variables. Only Yes/No target responses are eligible. The training script saves a per-column audit with missing percentages and cardinalities and produces five exploratory visualisations: target balance, missingness, province, geotype and drinking-water source. It also reports duplicate records among eligible cases. Numeric-looking survey category labels are handled as categories, avoiding unjustified ordering.

Data are split 60% training, 20% validation and 20% held-out test with target stratification and fixed seed 42. All one-hot encoding is fitted on the training split only. The target and direct interruption follow-up variables are excluded from predictors. Caution: this course demonstration does not implement PSU-aware splitting, survey-weighted population estimates or a fully documented missing-value-code treatment. These limitations affect generalisability.

3. Baseline and four substantive models (Question 3)

A majority-class DummyClassifier establishes a naïve reference. Traditional models: class-weighted Logistic Regression and Random Forest. Neural models: a dense multilayer perceptron (MLP) and a residual dense tabular network. Both neural models use binary cross-entropy with logits, AdamW, dropout, fixed seed, batches of approximately 256, a maximum of 35 epochs and validation-loss early stopping (patience 6). The residual tabular architecture is subject to lecturer approval as the second neural family.

4. Evaluation, comparison and overfitting (Question 4)

After running train.py, insert the actual values from outputs/validation_comparison.csv and outputs/test_comparison.csv. Attach outputs/confusion_*.png, classification_*.txt, roc_comparison.png, model_comparison.png and learning_*.png. Discuss differences between training and validation losses, whether class imbalance affects recall, and whether high accuracy masks poor minority-class performance. Do not claim cross-validation confidence intervals that have not been calculated.

5. Champion optimisation (Question 5)

The provisional champion is chosen using validation macro-F1 only. Traditional champions undergo a documented 3-fold randomized parameter search on the training partition. Neural champions undergo a validation-only dropout search. The script records the selected settings in outputs/tuning.json. The held-out test is reserved for a single final evaluation stage after model selection. Compare validation scores before and after tuning; describe practical differences separately from statistical significance.

6. Ablation and McNemar test (Question 6)

Three controlled feature-group removals assess province/geotype, drinking-water source, and housing/tenure. The resulting validation macro-F1 values are saved in outputs/ablations.csv. Note: neural ablations use a shortened training budget and are not strictly one-change-at-a-time comparisons; rerun with the same training schedule for full rubric compliance. For McNemar, H0: the two models have equal probabilities of a correct prediction on the same cases; H1: they differ. At alpha = 0.05, use the exact binomial p-value and the four correctness counts in outputs/mcnemar.json. Explain that statistical significance does not establish service benefit.

7. Deployment and responsible use (Question 7)

A Streamlit interface is provided in app.py. Run python train.py followed by streamlit run app.py. The app loads the saved full pipeline (or saved neural preprocessing plus network), presents validated category choices, reports the estimated probability and predicted class, and displays a responsible-use warning. For demonstration, record screenshots of three distinct valid households and an invalid case (for example, missing model artifacts before training). Do not expose household identifiers.

8. Recommendations, limitations and reproducibility (Question 8)

Use the model only as exploratory evidence alongside municipal maintenance logs, household engagement and validated local service records. Do not infer causality or make household-level entitlement decisions. Monitor class-wise recall, geographic disparities, temporal drift and data quality. The README specifies installation, execution order, source information and limitations. A final presentation and signed group contribution statement must be completed by the team.

References and acknowledgement

Statistics South Africa. General Household Survey 2023, household microdata (uploaded file; verify dataset-specific documentation and licence).

IDA117V. Introduction to Data Science: Group Project Template, Semester 2, 2026.

Libraries used in the supplied implementation: pandas, NumPy, scikit-learn, SciPy, Matplotlib, PyTorch, Streamlit and joblib. AI assistance was used to draft the code and report; disclose its use according to institutional rules.